Introduction to Computer Science

Lecure2_Part1

Computer components

OBJECTIVES AND INTENDED OUTCOMES

- By the end of this lesson the learner is expected to:
  - Understand the concept of processor and processing.
  - Learn the function of Central Processing Unit (CPU).
  - Learn the function of control unit, and Arithmetic and Logic unit.
  - Learn the compare between RAM and ROM.
  - Understand how a CPU works.
  - Understand the concept of the main storage unit and secondary storage units.
  - Learn about the different types of secondary storage units.
  - Learn about the storage.

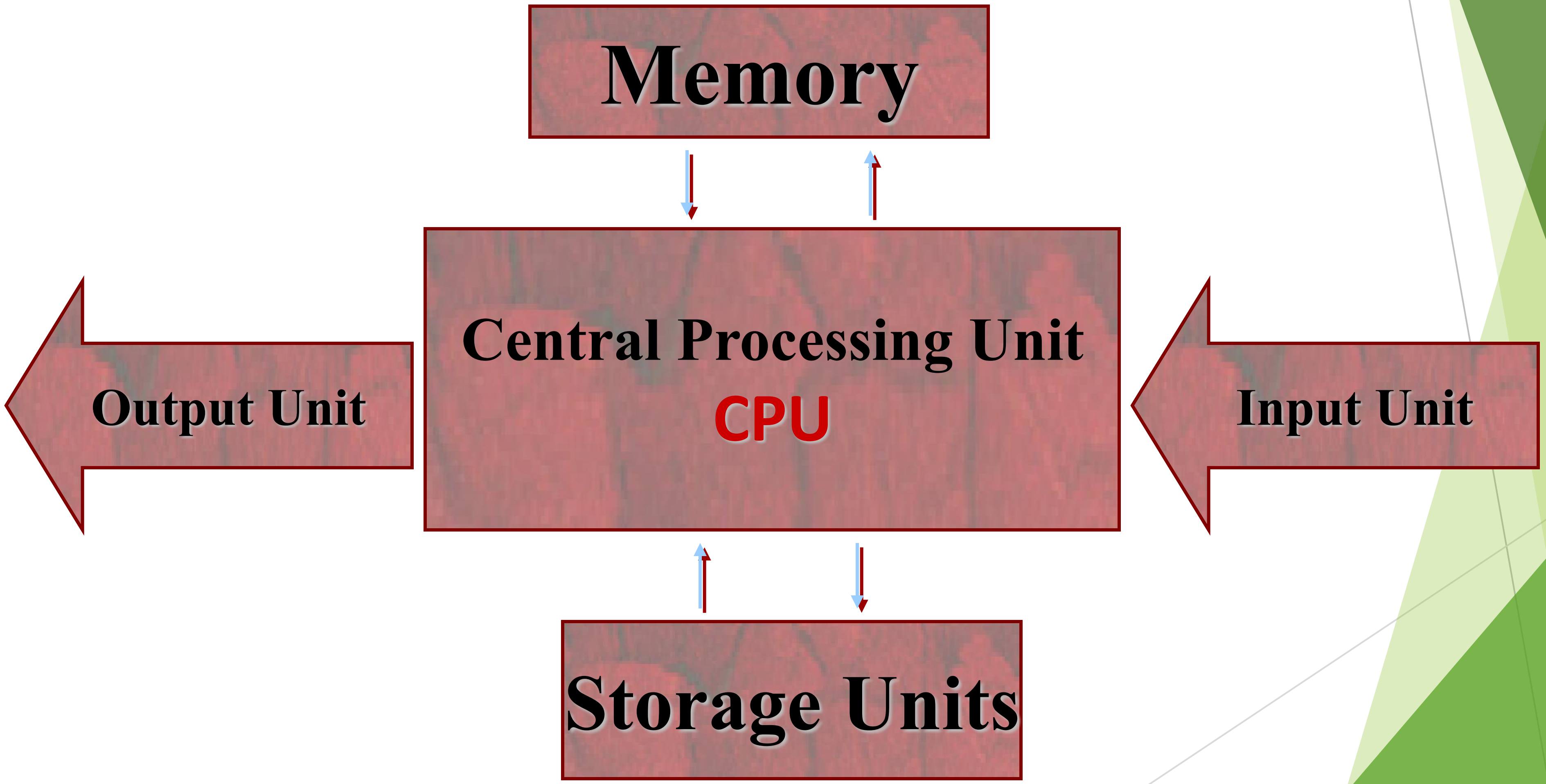
Computer components

Computer Components

Hardware

Software

Hardware


Memory
Central Processing Unit
CPU
Output Unit
Input Unit
Storage Units

Software

- ❑ divided into :
  - ❑ Systems software
  - ❑ Application software.

Central Processing Unit

- The main components of a computer. It is a chip consisting of a large number of electronic circuits made of silicon.
- Its tasks:
  - Data analysis.
  - Sending instructions to the rest of the computer.
  - Carry out instructions.
  - Control of inputs and outputs.

Central Processing Unit components

Central Processing Unit

Arithmetic and Logic (ALU)	Control unit **(CU)**
It performs all arithmetic and logical operations.	It controls all parts of the computer and delivers instructions from one part to another.

- The faster computer, the higher processor speed
- Computer speed is measured in megahertz (MHz).

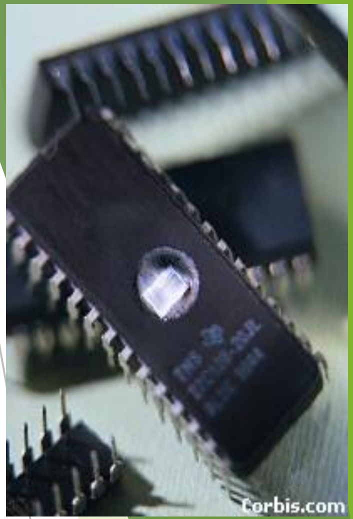
Central Processing Unit

- The processor capacity can be measured by the temporal speed: which refers to the number of operations that the processor performs per second and is measured in megahertz.
  - The processor communicates physically with other parts of the computer through three types of buses :
  - Data Buses: Transfer data across the various units of the computer
  - Addressing Buses : Transfer the addresses of the main memory cells specified in the processor
  - Control Buses: Transfer commands to the console

Arithmetic and Logic (ALU)

- The **ALU** needs data to process it and instructions that specify the type of processing required, and the **ALU** gets that from the main memory and uses the main memory in four basic activities:
  - Store the entered.
  - Store the interim results of operations.
  - Store the final results of the operation.
  - Store operations and operating orders.

Computer Memory



- Read Only Memory (ROM):
  - A computer memory that stores operating instructions for the device.
  - It is permanent and cannot be erased or written on.
  - set by the device manufacturer.
- Random Access Memory (RAM):
  - Memory starts with the computer running.
  - Temporary can be erased and written to.
  - Data is lost in the event of a power outage.
  - It is the mediator between the CPU and storage units.

Difference between RAM & ROM

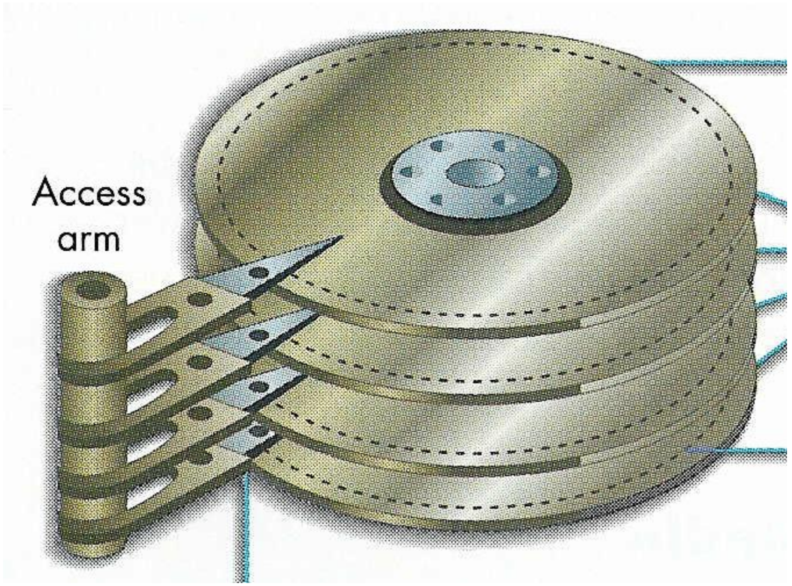
	ROM	RAM
Capacity	Limited storage capacity	More storage capacity
Storage Type	permanent	temporary
Contents	CPU instructions	programs and user data
Writing	Once when setting up the device	At what time
reading	every time you turn on the device	At what time

secondary storage units

- **Hard Disk:**
  - Huge storage capacity.
  - High speed access to the data stored in it.
  - It is usually installed in the device and may be connected from the outside.
- **Floppy Disk:**
  - Limited storage capacity of 1.4MB, low cost.
- **Flash Memory:**
  - It has the ability to connect externally.
  - With large storage capabilities.
  - High cost.
- **CD-ROM:**
  - The data is stored using a laser.
  - The storage capacity of DVD discs is up to a number of gigabytes.

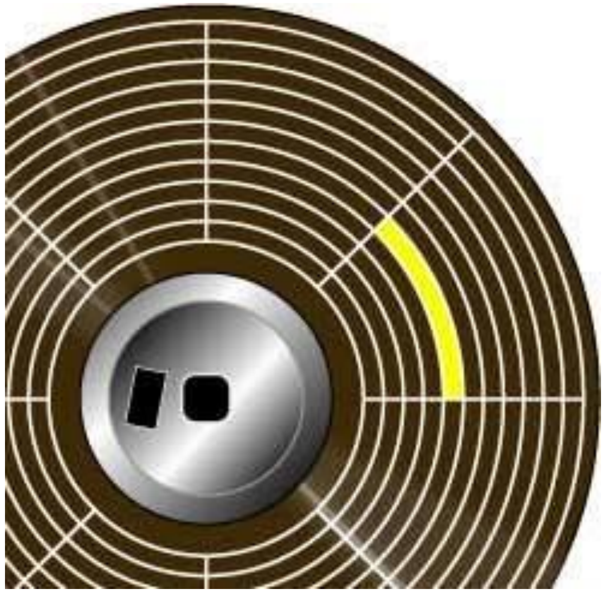
secondary storage

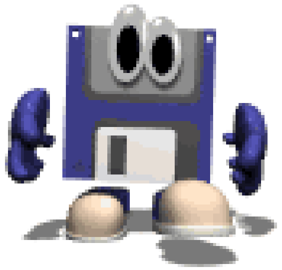
Hard Disk



- These discs are stored magnetically.

Floppy Disk





- In the past, it was the most used storage unit, now it has been replaced by other storage unit.
- It is symbolized by (A).

Optical Disk

- A technology that uses laser beams for storage.
- Now Read-Write discs.

DVD

- A technology that uses laser beams for storage.
- Its capacity is 6 times greater than the CD.

Flash Memory

Introduction to Computer Science

Lecure2_Part2

Computer components

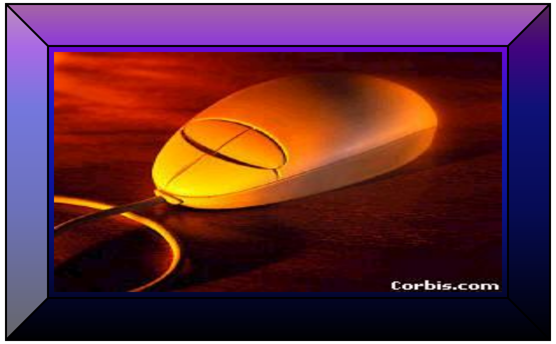
OBJECTIVES AND INTENDED OUTCOMES

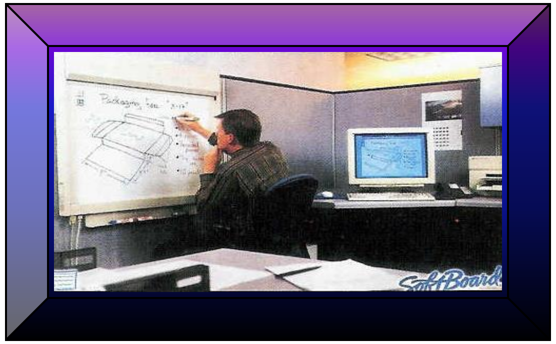
- By the end of this lesson the learner is expected to:
  - Understand the concept of the input and output devices.
  - Learn the function of input and output devices.
  - Learn the compare between deferent input and output devices.
  - Understand the appropriate way to enter and show various data.

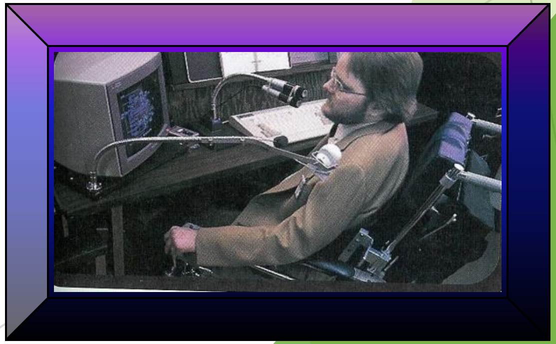
Input Unit

Input units are mainly based on linking computer users to the computers themselves, as they are responsible for entering data and programs into the computer's memory, and they send those entered data to the computer in a language readable by the computer itself.

Input Unit





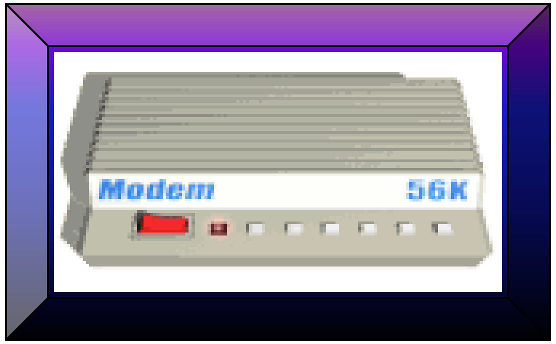


Output Unit

The output units that receive the results and display them in different forms.

Output Unit



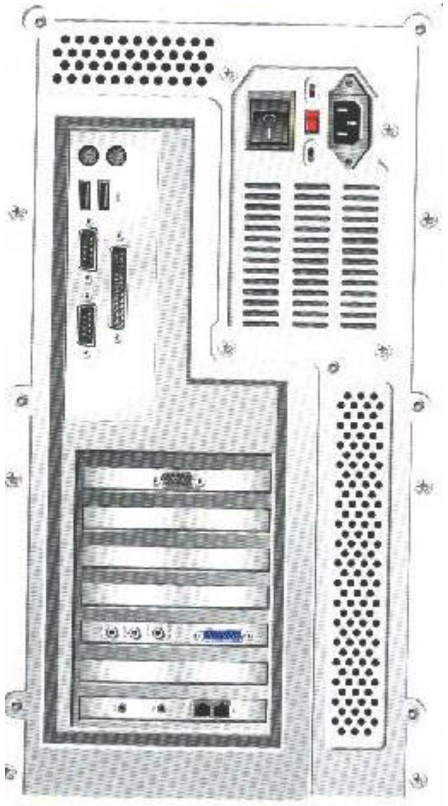




system Box

- The system box contains most of the computer's units.
- Connect to other computer units. Its most important parts:
  - Motherboard
  - Ports
  - Power Supply

MotherBoard

Ports

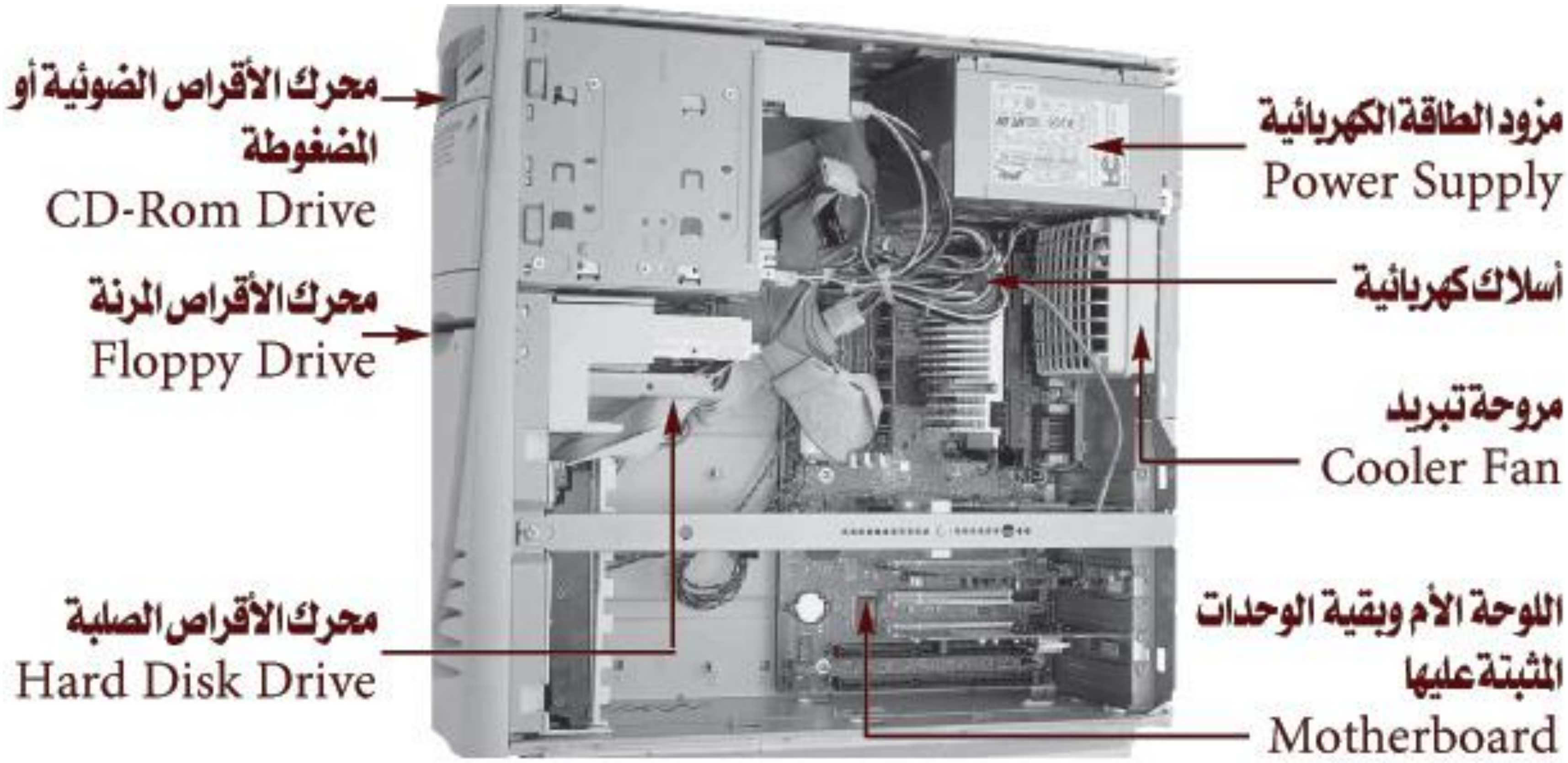


- Parallel ports > Printer, Serial ports > Mouse, USB ports.
- Expansion ports.

Power Supply

- Supplying the computer with electricity.
- There is a fan to reduce the heat of the generator.

HW Components



HW Components